Theory of Elementary Particles
Lecture 9

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Euler-Lagrange Equations of Motions

Let us consider action S which is given by the integral

$$S = \int_V d^4x \mathcal{L}(..). \tag{1}$$

$\mathcal{L}(..)$ is density of the Lagrangian.

- Let action depends on scalar field $\phi(x)$ and its first derivatives $\partial_\mu \phi(x)$.

We search for the configuration of fields which extremizes the action, hence

$$
\begin{aligned}
0 = \delta S &= \int_V d^4x \delta \mathcal{L}(\phi, \partial_\mu \phi) && (2)\\
&= \int_V d^4x \left[\frac{\delta \mathcal{L}}{\delta \phi} \delta\phi + \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} \delta \partial_\mu \phi \right] && (3)\\
&= \int_V d^4x \left[\frac{\delta \mathcal{L}}{\delta \phi} \delta\phi + \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} \partial_\mu \delta \phi \right] && (4)
\end{aligned}
$$

where we assumed that

$$\partial_\mu \delta\phi = \delta \partial_\mu \phi. \quad (5)$$

Then

$$\delta S = \int_V d^4x \left[\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} \right] \delta\phi + \int_V d^4x \partial_\mu \left[\frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} \delta\phi \right] \quad (6)$$

We have

$$
\begin{aligned}
* \quad &\equiv \quad \int_V d^4x \partial_\mu \left[\frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} \delta\phi \right] && (7) \\
&= \quad \int_{t_i}^{t_f} dt \int_V d^3x \partial_0 \left[\frac{\delta \mathcal{L}}{\delta \partial_0 \phi} \delta\phi \right] + \int_{t_i}^{t_f} dt \int_V d^3x \sum_{i=1}^{3} \partial_i \left[\frac{\delta \mathcal{L}}{\delta \partial_i \phi} \delta\phi \right] && (8) \\
&= \quad \int_V d^3x \left[\frac{\delta \mathcal{L}}{\delta \partial_0 \phi} \delta\phi \right] \Big|_{t_i}^{t_f} + \int_{t_i}^{t_f} dt \int_{\partial V} d\vec{S} \left[\frac{\delta \mathcal{L}}{\delta(\nabla \phi)} \delta\phi \right] && (9)
\end{aligned}
$$

Typically it is assumed that the initial and the final configurations of fields are known. It leads to Dirichlet boundary conditions:

$$
\delta\phi|_{t=t_i,\partial V} = 0, \quad \delta\phi|_{t=t_f,\partial V} = 0 \tag{10}
$$

we get **Euler-Lagrange Equations**:

$$
0 = \frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi}. \tag{11}
$$

Notice that it can be also assumed that:

$$
0 = \left. \frac{\delta \mathcal{L}}{\delta \partial_0 \phi} \right|_{t=t_i} = \left. \frac{\delta \mathcal{L}}{\delta \partial_0 \phi} \right|_{t=t_f} = \left. \frac{\delta \mathcal{L}}{\delta(\nabla \phi)} \right|_{\partial V}. \tag{12}
$$

Noether's Theorem

- One can distinguish the transformations which remain action invariant. In this case the density of the Lagrangian may transform as

$$\mathcal{L} \to \mathcal{L} + \partial^{\mu} J_{\mu}. \tag{13}$$

- or investigate how density of the Lagrangian transforms with respect of symmetry group G

$$\delta\mathcal{L} = (E - L\ equations)\,\delta\phi + \partial_{\mu} j^{\mu}, \tag{14}$$

where

$$j^{\mu} = \frac{\delta\mathcal{L}}{\delta\partial_{\mu}\phi}\delta\phi. \tag{15}$$

$\delta\phi$ is given by symmetry group.

- From the invariance we have that

$$\partial_{\mu}\left(j^{\mu} - J^{\mu}\right) = 0 \tag{16}$$

so the conserved current, called Noether current, reads

$$N_{\mu} = j_{\mu} - J_{\mu}. \tag{17}$$

Remark

Notice that if density of the lagrangian is invariant to given symmetry transformations then

$$N^{\mu} = j^{\mu},$$

which is typical for global symmetries.

Noether Charges

Conserved currents give rise to conserved charges:

$$Q^i = \int d^3x\, j_i^0(x). \tag{18}$$

Example: Global Lie Group Symmetry

Consider a global symmetry generated by T_i, $i = 1, \ldots, M$:

$$\phi'(x) = \exp(-i\theta_i T_i)\phi(x). \tag{19}$$

For an infinitesimal transformation,

$$\delta\phi(x) = -i\theta_i T_i \phi(x). \tag{20}$$

For a Lagrangian invariant under this transformation, the Noether currents read

$$j_i^\mu = -i\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} T_i \phi. \tag{21}$$

The corresponding charges are

$$Q^i = \int d^3x\, j_i^0, \qquad \frac{dQ^i}{dt} = 0, \tag{22}$$

assuming vanishing current flux at infinity.

Isospin symmetry case

- Let us consider $SU(2)$ group with elements $\exp(-i\theta_a T_a)$, $a = 1, 2, 3$, $T_a = \tau_a/2$, note that $[T_a, T_b] = i\epsilon_{abc}T_c$
- The density of the lagrangian for isodublet of nucleon fields reads

$$\mathcal{L}(x) = \bar{\Psi}_N(x)(i\gamma^\mu \partial_\mu - m)\Psi_N(x), \tag{23}$$

where

$$\Psi = \begin{pmatrix} \psi_p \\ \psi_n \end{pmatrix} \tag{24}$$

We compute the Noether currents

$$j_i^\mu = \frac{\delta \mathcal{L}}{\delta\, \partial_\mu \Psi_N} \delta_i \Psi_N = i\overline{\Psi}_N \gamma^\mu \cdot (-iT_i \Psi_N) = \overline{\Psi}_N(x)\gamma^\mu T_i \Psi_N \tag{25}$$

and $J^\mu = 0$. Then the conserved charge reads

$$Q^i = \int d^3x j_0^i = \int d^3x \Psi_N^\dagger(x) T_i \Psi_N(x). \tag{26}$$

Energy-Momentum Tensor

- Consider infinitesimal variation of the space and the time coordinates:

$$x^\mu \to x^\mu + a^\mu, \quad \phi(x+a) \approx \phi(x) + a^\mu \partial_\mu \phi(x), \quad \Rightarrow \quad \delta\phi = a^\mu \partial_\mu \phi(x). \tag{27}$$

For the density of the Lagrangian we get

$$\delta\mathcal{L} \approx a^\mu \partial_\mu \mathcal{L} \tag{28}$$

The total conserved current reads

$$N^\mu = j^\mu - J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} a^\nu \partial_\nu \phi - a^\mu \mathcal{L} = a^\nu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - g_\nu{}^\mu \mathcal{L} \right) \tag{29}$$

$T^\mu{}_\nu$ is called energy momentum tensor and reads:

$$T^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - g^\mu_\nu \mathcal{L}. \tag{30}$$

Component 00 gives Hamiltonian (energy)

$$T^{00} = \pi\, \partial_0 \phi - \mathcal{L} = \mathcal{H}, \tag{31}$$

where the canonical momentum is defined by:

$$\pi = \frac{\partial \mathcal{L}}{\partial \partial_0 \phi}, \tag{32}$$

as well as

$$H = \int d^3x \mathcal{H}. \tag{33}$$

Conserved momentum will be given by

$$T^{0i} = \pi\, \partial^i \phi = -\pi\, \nabla^i \phi. \tag{34}$$

indeed

$$\vec{P} = -\int d^3x \pi\, \nabla \phi \tag{35}$$

Classical QED

- Let us consider the free photon field A_μ.
- The Maxwell equations are obtained from the theory described by the density of the Lagrangian

$$\mathcal{L}_{EM}(x) = -\frac{1}{4} F_{\mu\nu}(x) F^{\mu\nu}(x) - j_\mu(x) A^\mu(x), \tag{36}$$

where

$$F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x), \tag{37}$$

$$j^\mu = (\rho, \mathbf{j}).$$

The E-M (equations of motion), Maxwell equations, of classical electrodynamics read

$$\partial_\mu F^{\mu\nu} = j^\nu \tag{38}$$

Current conservation is the result of Maxwell equations,

$$\partial_\alpha j^\alpha = \partial_\nu \partial_\mu F^{\mu\nu} = 0 \tag{39}$$

Gauge Invariance

If the current is conserved it can be easily shown that the Lagrangian

$$\int_V d^4x \mathcal{L}_{EM}(x)$$

is invariant under the local gauge transformation,

$$A_\mu(x) \to A_\mu(x) - \partial_\mu \lambda(x) \tag{40}$$

where $\lambda(x)$ is arbitrary real function. Indeed,

$$F_{\mu\nu} \to F_{\mu\nu} \tag{41}$$

$$j^\mu(x)A_\mu(x) \to j^\mu(x)A_\mu(x) - j^\mu(x)\partial_\mu\lambda(x) \tag{42}$$

But

$$\int_V d^4x j_\mu(x)\partial^\mu\lambda(x) = -\int d^4x \partial^\mu j_\mu(x)\lambda(x) = 0 \tag{43}$$

if

$$\int_{\partial V} ds^\mu j_\mu(x)\lambda(x) = 0. \tag{44}$$

Quantization

- In the canonical quantization, we consider the generalized position operators, in this case A^μ, and canonical momenta, in this case,

$$\pi^\mu(x) = \frac{\partial \mathcal{L}_{EM}(x)}{\partial \dot{A}_\mu(x)} \tag{45}$$

- Notice that the field $\dot{A}^0$ does not appear in the Lagrangian. Hence π^0 vanishes and the commutator $[A^0, \pi^0]$ is also zero. But it should take the form

$$[A^0(\mathbf{x}), \pi^0(\mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

This property of the QED complicates the canonical quantization.

With a help of the gauge invariance (which allows to re-define the fields) one can eliminate the "problem of A^0 component". There are two general attempts:

(A) eliminating A^0 field by expressing it by the other components of A^μ.

(B) adding additional term to the Lagrangian containing $\partial_0 A_0$ field.

Both ideas can be realized if one uses the local gauge symmetry satisfied by the QED theory.

Two typical gauge choices,

- Coulomb gauge

$$\nabla \cdot \mathbf{A} = 0 \tag{46}$$

  it is typically used for the studies of the atomic and other low energy systems. In this case the case (A) of the quantization is naturally considered.

- Lorenz gauge

$$\partial_\mu A^\mu = 0 \tag{47}$$

  it is typically chosen for discussing high-energy scattering processes. In the case (B) the quantization can be naturally performed.

Coulomb gauge

For the Coulomb gauge A^0 field satisfies the Poisson's equation,

$$j^0 \equiv \rho = \partial_\mu F^{\mu 0} = \Box A^0 - \partial_0 \partial_0 A^0 - \partial^0 \nabla \cdot \mathbf{A} = -\nabla^2 A^0 \tag{48}$$

The solution is given by the Coulomb's law,

$$A^0(\mathbf{r}) = \frac{1}{4\pi} \int d^3 r' \frac{\rho(\mathbf{r}')}{|\mathbf{r}' - \mathbf{r}|} \tag{49}$$

It is an instantaneous (non-retarded) solution!

Lorenz Gauge

- Consider one of Maxwell's equations:

$$j^0 \equiv \rho = \partial_\mu F^{\mu 0} = \Box A^0 - \partial^0(\partial_\mu A^\mu). \tag{50}$$

- In the Lorenz gauge,

$$\partial_\mu A^\mu = 0, \tag{51}$$

the equation reduces to

$$\Box A^0 = \rho. \tag{52}$$

This is a wave equation with a source.

- To fix the gauge in the Lagrangian, we introduce the covariant gauge-fixing term

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2, \tag{53}$$

where ξ is a gauge parameter.
  - $\xi = 1$: Feynman gauge,
  - $\xi \to 0$: Landau gauge,
  - $\xi > 0$: covariant R_ξ gauges.

- The most general form of the electromagnetic Lagrangian for E-M field reads

$$\mathcal{L}_{QED-possibility}(x) = \lambda_1 F_{\mu\nu}F^{\mu\nu} + \lambda_2 G_{\mu\nu}G^{\mu\nu} + M_\gamma^2 A^\mu A_\mu, \qquad (54)$$

  where $G_{\mu\nu} = \partial_\mu A_\nu + \partial_\nu A_\mu$.
- It can be easily seen that after the gauge transformation,

$$G_{\mu\nu} \to G_{\mu\nu} - 2\partial_\mu\partial_\nu\lambda \qquad (55)$$

  In order to be gauge invariant we must set $\lambda_2 = 0$
- The mass term,

$$A_\mu A^\mu \to A_\mu A^\mu - 2A^\mu\partial_\mu\lambda + \partial_\mu\lambda\partial^\mu\lambda. \qquad (56)$$

  hence $M_\gamma = 0$.

QED Lagrangian

- Now let us introduce the density of the QED Lagrangian containing the electron and photon fields,

$$\mathcal{L}_{QED}(x) = \underbrace{\bar{\psi}(x)(i\gamma^\mu\partial_\mu - m)\psi(x)}_{Dirac} \underbrace{-\frac{1}{4}F_{\mu\nu}(x)F^{\mu\nu}(x)}_{Maxwell} \underbrace{-e\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)}_{Interaction}, \tag{57}$$

$$F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) \tag{58}$$

- We keep the normalization $e = -|e|$; for a fermion of charge $Q|e|$ $(Q > 0)$ replace $e \to -Qe$ (cf. Lecture 2).
- The equations of motions

$$(i\hat{\partial} - m)\psi(x) = e\gamma^\mu A_\mu(x)\psi(x) \tag{59}$$

$$(i\gamma^\mu(\partial_\mu + ieA_\mu(x)) - m)\psi(x) = 0 \tag{60}$$

and

$$\partial_\mu F^{\mu\nu}(x) = e\bar{\psi}(x)\gamma^\nu\psi(x) = e\mathcal{J}^\nu(x) \quad \Rightarrow \quad \Box A^\nu(x) - \partial^\nu\partial_\mu A^\mu = e\mathcal{J}^\nu(x) \tag{61}$$

Covariant Derivative

- Let us introduce the covariant derivative,

$$D_\mu \equiv \partial_\mu + ieA_\mu(x) \tag{62}$$

- The QED is invariant to:
  - global $U(1)$ gauge symmetry,

$$\psi \to e^{i\lambda}\psi$$

  - local $U(1)$ gauge symmetry

$$\psi(x) \to e^{i\lambda(x)}\psi(x), \quad A_\mu \to A'_\mu \equiv A_\mu - \frac{1}{e}\partial_\mu\lambda(x) \tag{63}$$

Indeed: $\psi' = e^{i\lambda(x)}\psi$

$$\begin{aligned}
(D_\mu\psi)' &= D'_\mu\psi' = (\partial_\mu + ieA'_\mu(x))e^{i\lambda(x)}\psi && (64)\\
&= e^{i\lambda(x)}(\partial_\mu + i\partial_\mu\lambda(x) + ieA'_\mu(x))\psi && (65)\\
&= e^{i\lambda(x)}(\partial_\mu + ieA_\mu(x))\psi && (66)\\
&= e^{i\lambda(x)}D_\mu\psi && (67)
\end{aligned}$$

Now let us calculate

$$D_\mu D_\nu = (\partial_\mu + ieA_\mu(x))(\partial_\nu + ieA_\nu(x)) \quad (68)$$

$$= \underbrace{\partial_\mu\partial_\nu - e^2 A_\mu(x)A_\nu(x) + ieA_\mu(x)\partial_\nu + ieA_\nu(x)\partial_\mu}_{Sym.} + ie\partial_\mu A_\nu(x) \quad (69)$$

We see that

$$(D_\mu D_\nu - D_\nu D_\mu) \equiv ieF_{\mu\nu} \quad (70)$$

and $F'_{\mu\nu} = F_{\mu\nu}$

- The photon is massless because $A_\mu A^\mu$ term is not gauge invariant.
- The minimal coupling of photon with electron is contained in the covariant derivative, which can be constructed from the transformation property of the electron field.
- QED Lagrangian does not contain the self-coupling photon terms, and photons do not carry charge

Ghosts

Consider QED in a covariant gauge.

- The gauge-fixed QED Lagrangian reads

$$\mathcal{L}_{QED} = \bar{\psi}(i\hat{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2. \tag{71}$$

- Under the gauge transformation

$$A_\mu \to A_\mu - \frac{1}{e}\partial_\mu \lambda, \tag{72}$$

the gauge-fixing function transforms as

$$\partial_\mu A^\mu \to \partial_\mu A^\mu - \frac{1}{e}\Box\lambda. \tag{73}$$

- The Lorenz gauge condition is preserved by residual transformations satisfying

$$\Box\lambda = 0. \tag{74}$$

- In QED, the Faddeev–Popov operator is independent of A_μ. Hence the ghosts are free and decouple.
- In non-Abelian Yang–Mills theory, the Faddeev-Popov operator depends on the gauge field, and ghosts interact with gauge bosons.

- Notice that in the covariant gauge $\partial^\mu A_\mu = 0$ the equation of motion of free photon field reads,

$$\Box A^\nu(x) = 0, \tag{75}$$

  but it is gauge invariant only if $\Box\lambda(x) = 0$.
- Specifying $\lambda(x)$ leads to particular realization for the photon field and its photon propagator in the Lorenz gauge.

Yang-Mills Theory

- Consider $SU(N)$ invariant model described by lagrangian:

$$\mathcal{L} = \sum_{k=1}^{N} \bar{\psi}_k(x)(i\gamma^\mu \partial_\mu - m)\psi_k(x) \tag{76}$$

where

$$\Psi = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix}$$

- Global symmetry:

$$\Psi \to U\Psi \tag{77}$$

$U \in SU(N)$, where

$$U = \exp(-iT^a\theta^a), \tag{78}$$

where

$$[T^a, T^b] = if^{abc}T^c$$

- $U \to U(x)$, where $U(x) = \exp(-iT^a\theta^a(x))$
-

$$\bar{\Psi}(x)\partial_\mu\Psi(x) \to \bar{\Psi}(x)\partial_\mu\Psi(x) + \bar{\Psi}(x)U^{-1}\partial_\mu U\Psi(x) \tag{79}$$

- We introduce the covariant derivative (we sum over repeated indexes e.g. a's)

$$D_\mu\Psi = \left(\partial_\mu + igT^a A^a_\mu\right)\Psi \tag{80}$$

- We demand that

$$(D_\mu\Psi)' = D'_\mu\Psi' \tag{81}$$

so

$$UD_\mu\Psi = D'_\mu U\Psi \tag{82}$$

$$U(\partial_\mu + igT^a A^a_\mu)(\dots) = U\partial_\mu + \partial_\mu U + igT^a A'^a_\mu U(\dots) \quad | \cdot U^\dagger \tag{83}$$

$$\partial_\mu + U\partial_\mu U^\dagger + igUT^a U^\dagger A^a_\mu = \partial_\mu + U\partial_\mu U^\dagger + \partial_\mu UU^\dagger + igT^a A'^a_\mu \tag{84}$$

$$igUT^a U^\dagger A^a_\mu - \partial_\mu UU^\dagger = igT^a A'^a_\mu \tag{85}$$

$$UT^a U^\dagger A^a_\mu + \frac{i}{g}\partial_\mu UU^\dagger = T^a A'^a_\mu \tag{86}$$

Notice that

$$UT^a U^\dagger \approx (1 - iT^b\theta^b)T^a(1 + iT^b\theta^b) \tag{87}$$

$$\approx T^a - i[T^b, T^a]\theta^b \tag{88}$$

$$= T^a + f^{bac}T^c\theta^b \tag{89}$$

$$= T^a - f^{abc}T^c\theta^b \tag{90}$$

and

$$\partial_\mu UU^\dagger = \partial_\mu(-iT^a\theta^a)UU^\dagger = -iT^a\partial_\mu\theta^a \tag{91}$$

$$A'^{a}_{\mu} = A^{a}_{\mu} + f^{abc}\theta^{b}A^{c}_{\mu} + \frac{1}{g}\partial_{\mu}\theta^{a} \tag{92}$$

We see that A^{a}_{μ} transforms in the adjoint representation of $SU(N)$

- In analogy with QED we construct the second rank antisymmetric tensor

$$(D_\mu D_\nu - D_\nu D_\mu)\Psi \equiv igT^a F^a_{\mu\nu}\Psi \tag{93}$$

- $D_\mu\Psi$ has the same gauge transformation property as Ψ hence

$$\begin{aligned}((D_\mu D_\nu - D_\nu D_\mu)\Psi)' &= D'_\mu D'_\nu \Psi' - D'_\nu D'_\mu \Psi' && (94)\\ &= D'_\mu D'_\nu U\Psi - D'_\nu D'_\mu U\Psi && (95)\\ &= D'_\mu U D_\nu \Psi - D'_\nu U D_\mu \Psi && (96)\\ &= U(D_\mu D_\nu - D_\nu D_\mu)\Psi && (97)\end{aligned}$$

$$
\begin{aligned}
D_\mu D_\nu \Psi &= (\partial_\mu + igA^a_\mu T^a)(\partial_\nu + igA^b_\nu T^b)\Psi \\
&= \left(\underbrace{\partial_\mu \partial_\nu}_{Sym.} + ig\partial_\mu A^b_\nu T^b + \underbrace{igA^b_\nu T^b \partial_\mu + igA^a_\mu T^a \partial_\nu}_{Sym.} - g^2 A^a_\mu A^b_\nu T^a T^b \right) \Psi
\end{aligned}
$$

Hence

$$
\begin{aligned}
(D_\mu D_\nu - D_\nu D_\mu) &= ig(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)T^a + ig(ig)[A^a_\mu T^a, A^b_\nu T^b] && (98)\\
&= ig(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)T^a + ig(ig)if^{cab}T^c A^a_\mu A^b_\nu && (99)\\
&= ig(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)T^a - ig^2 f^{cab}T^c A^a_\mu A^b_\nu && (100)\\
&= igF^a_{\mu\nu}T^a && (101)
\end{aligned}
$$

Hence

$$
F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu - gf^{abc}A^b_\mu A^c_\nu \tag{102}
$$

or

$$
\mathbf{F}_{\mu\nu} = \partial_\mu \mathbf{A}_\nu - \partial_\nu \mathbf{A}_\mu + ig[\mathbf{A}_\mu, \mathbf{A}_\nu] \tag{103}
$$

Notice that

$$
\mathbf{F'}_{\mu\nu} U\Psi = \mathbf{F}_{\mu\nu} U\Psi \tag{104}
$$

hence

$$
\mathbf{F'}_{\mu\nu} = U\mathbf{F}_{\mu\nu}U^\dagger \approx \mathbf{F}_{\mu\nu} - i[\boldsymbol{\theta}, \mathbf{F}_{\mu\nu}] \tag{105}
$$

We see that $F^a_{\mu\nu}$ transforms in the adjoint representation of $SU(N)$

$$F^a_{\mu\nu}F^{a\mu\nu} = (\partial_\mu A^a_\nu - \partial_\nu A^a_\mu - g f^{abc} A^b_\mu A^c_\nu)(\partial^\mu A^{a\nu} - \partial^\nu A^{a\mu} - g f^{abc} A^{b\mu} A^{c\nu}) \quad (106)$$

$$= (\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)(\partial^\mu A^{a\nu} - \partial^\nu A^{a\mu}) \quad (107)$$

$$-2g f^{abc}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) A^{b\mu} A^{c\nu} + g^2 f^{abc} f^{ade} A^b_\mu A^c_\nu A^{d\mu} A^{e\nu} \quad (108)$$

$$-\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} = \mathcal{L}_0 + \mathcal{L}_{AAA}(x) + \mathcal{L}_{AAAA}(x) \quad (109)$$

$$\mathcal{L}_{AAA}(x) = \frac{1}{2} g f^{abc}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) A^{b\mu} A^{c\nu} \quad (110)$$

$$\mathcal{L}_{AAAA}(x) = -\frac{g^2}{4} f^{abc} f^{ade} A^b_\mu A^c_\nu A^{d\mu} A^{e\nu} \quad (111)$$

- We notice that

$$A^a_\mu(q) = \frac{1}{(2\pi)^4}\int d^4x A^a_\mu(x) e^{-iqx} \tag{112}$$

Since f^{abc} is totally antisymmetric,

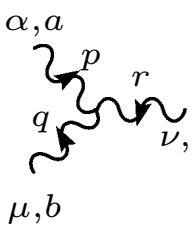
$$\begin{aligned}
* &= i\int d^4x \mathcal{L}_{AAA}(x) \qquad (113)\\
&= -\frac{1}{2}\frac{gf^{abc}}{(2\pi)^{12}}\int d^4x\, d^4p\, d^4q\, d^4r\, (p_\mu A^a_\nu(p) - p_\nu A^a_\mu(p)) A^{b\mu}(q) A^{c\nu}(r) e^{ix(p+q+r)}\\
&= -\frac{1}{2}\frac{gf^{abc}}{(2\pi)^{8}}\int d^4p\, d^4q\, d^4r\, (p_\mu A^a_\nu(p) - p_\nu A^a_\mu(p)) A^{b\mu}(q) A^{c\nu}(r) \delta^{(4)}(p+q+r)\\
&= -\frac{1}{2}\frac{gf^{abc}}{(2\pi)^{8}}\int d^4p\, d^4q\, d^4r\, (p_\mu g_{\alpha\nu} - p_\nu g_{\alpha\mu}) A^{a\alpha}(p) A^{b\mu}(q) A^{c\nu}(r) \delta^{(4)}(p+q+r)
\end{aligned}$$

We interchange $A^{a\alpha}(p)$ with $A^{b\mu}(q)$ (-1) permutation sign $f^{abc} = -f^{bac}$, and $A^{a\alpha}$ with $A^{c\nu}$, $f^{bac} = -f^{bca}$

$$\begin{aligned}
* &= -\frac{1}{2\cdot 3}\frac{gf^{abc}}{(2\pi)^{8}}\int d^4p\, d^4q\, d^4r\, (p_\mu g_{\alpha\nu} - p_\nu g_{\alpha\mu} - q_\alpha g_{\mu\nu} + q_\nu g_{\mu\alpha} - r_\mu g_{\nu\alpha} + r_\alpha g_{\nu}\\
&\quad A^{a\alpha}(p) A^{b\mu}(q) A^{c\nu}(r) \delta^{(4)}(p+q+r) \qquad (11
\end{aligned}$$

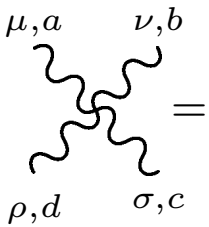
$$
\begin{aligned}
* \quad &= \quad -\frac{1}{2\cdot 3}\frac{g f^{abc}}{(2\pi)^8}\int d^4p\, d^4q\, d^4r\, (p_\mu g_{\alpha\nu} - p_\nu g_{\alpha\mu} - q_\alpha g_{\mu\nu} + q_\nu g_{\mu\alpha} - r_\mu g_{\nu\alpha} + r_\alpha g_{\nu\mu}) \\
&\quad A^{a\alpha}(p)A^{b\mu}(q)A^{c\nu}(r)\delta^{(4)}(p+q+r) \qquad (115) \\
&= \quad -\frac{1}{2\cdot 3}\frac{g f^{abc}}{(2\pi)^8}\int d^4p\, d^4q\, d^4r\, (g_{\alpha\nu}(p_\mu - r_\mu) + g_{\alpha\mu}(q_\nu - p_\nu) + g_{\nu\mu}(r_\alpha - q_\alpha)) \\
&\quad A^{a\alpha}(p)A^{b\mu}(q)A^{c\nu}(r)\delta^{(4)}(p+q+r) \qquad (116)
\end{aligned}
$$

Hence



$$= g f^{abc}(g_{\alpha\nu}(r_\mu - p_\mu) + g_{\alpha\mu}(p_\nu - q_\nu) + g_{\nu\mu}(q_\alpha - r_\alpha)) \qquad (117)$$

Analogously



$$= -ig^2 \Big(f^{abe} f^{cde} (g_{\mu\sigma} g_{\nu\rho} - g_{\mu\rho} g_{\nu\sigma}) + f^{ace} f^{bde} (g_{\mu\nu} g_{\sigma\rho} - g_{\mu\rho} g_{\nu\sigma}) + f^{ade} f^{cbe} (g_{\mu\sigma} g_{\nu\rho} - g_{\mu\nu} g_{\rho\sigma}) \Big)$$

- A theory which is invariant under the local $SU(N)$ gauge group is referred to as Yang-Mills theory, e.g. Quantum Chromodynamics with $SU(3)$
- $$\mathcal{L}_{YM}(x) = \bar{\Psi}(x)\left(i\gamma^{\mu}D_{\mu} - m\right)\Psi(x) - \frac{1}{4}F^{a}_{\mu\nu}(x)F^{a,\mu\nu}(x) \tag{118}$$
- $N^2 - 1$ massless gauge bosons A^a_μ, which carry the group charge (adjoint representation); the Aff coupling reads
$$\mathcal{L}_{ffA}(x) = -g\bar{\Psi}\gamma^{\mu}T^{a}\Psi(x)A^{a}_{\mu}(x) \tag{119}$$
- Three gauge bosons coupling: AAA:
$$\mathcal{L}_{AAA}(x) = \frac{1}{2}gf^{abc}(\partial_{\mu}A^{a}_{\nu} - \partial_{\nu}A^{a}_{\mu})A^{b\mu}A^{c\nu} \tag{120}$$
- Four gauge bosons coupling: $AAAA$
$$\mathcal{L}_{AAAA}(x) = -\frac{g^2}{4}f^{abc}f^{ade}A^{b}_{\mu}A^{c}_{\nu}A^{d\mu}A^{e\nu} \tag{121}$$

Quantum Chromodynamics

- QCD is Yang-Mills theory with $SU(3)$ local gauge invariance.
- Fermions are called quarks
- Each fermion of given flavor has three internal degrees of freedom called color.
- There are 8 vector gauge bosons called gluons.

CP violation in QCD

- From the second rank tensor $F^a_{\mu\nu}$ we may easily construct dual tensor

$$\tilde{F}^a_{\mu\nu} = -\frac{1}{2}\epsilon_{\mu\nu\alpha\beta}F^{a,\alpha\beta} \tag{122}$$

- This tensor transforms under $SU(N)$ in the same way as $F^a_{\mu\nu}$
- Hence, one can add to QCD lagrangian the term (analog in QED of $\mathbf{E}\cdot\mathbf{B}$ interaction)

$$\mathcal{L}_\theta = \frac{\theta g^2}{32\pi^2}F^a_{\mu\nu}\tilde{F}^{a\mu\nu} \tag{123}$$

The normalization is chosen conventionally.

$\mathcal{P}$ and $\mathcal{C}$ Symmetries

Consider the QCD θ term:

$$\mathcal{L}_\theta \propto F^a_{\mu\nu}\widetilde{F}^{a\mu\nu}, \qquad \widetilde{F}^{a\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F^a_{\rho\sigma}. \tag{124}$$

- Under parity $\mathcal{P}$:

$$\begin{aligned} F^a_{0k}(t,\mathbf{x}) &\to -F^a_{0k}(t,-\mathbf{x}), \\ F^a_{km}(t,\mathbf{x}) &\to F^a_{km}(t,-\mathbf{x}). \end{aligned} \tag{125}$$

  Consequently,

$$F^a_{\mu\nu}\widetilde{F}^{a\mu\nu} \xrightarrow{\mathcal{P}} -F^a_{\mu\nu}\widetilde{F}^{a\mu\nu}, \tag{126}$$

  where the fields on the right are evaluated at the parity-reflected point.

- Under charge conjugation $\mathcal{C}$, in matrix notation $A_\mu = A^a_\mu T^a$:

$$A_\mu \xrightarrow{\mathcal{C}} -A^T_\mu, \qquad F_{\mu\nu} \xrightarrow{\mathcal{C}} -F^T_{\mu\nu}. \tag{127}$$

- Therefore,

$$\mathrm{Tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}) \quad \text{is } \mathcal{C}\text{-even and } \mathcal{P}\text{-odd.} \tag{128}$$

- Hence a nonzero θ term violates $\mathcal{P}$ and $\mathcal{CP}$.

$$\tilde{F}^a_{\alpha\beta} = -\frac{1}{2}\epsilon_{\alpha\beta\mu\nu}(\partial^\mu A^{a\nu} - \partial^\nu A^{a\mu} - g f^{abc} A^{b\mu} A^{c\nu}) \quad (129)$$

$$= -\epsilon_{\alpha\beta\mu\nu}\partial^\mu A^{a\nu} + \frac{g}{2} f^{abc}\epsilon_{\alpha\beta\mu\nu} A^{b\mu} A^{c\nu} \quad (130)$$

$$\tilde{F}^a_{\alpha\beta} F^{a\alpha\beta} = -2\epsilon_{\alpha\beta\mu\nu}\partial^\mu A^{a\nu}\partial^\alpha A^{a\beta} + g f^{abc}\epsilon_{\alpha\beta\mu\nu}\partial^\alpha A^{a\beta} A^{b\mu} A^{c\nu}$$

$$+ g f^{abc}\epsilon_{\alpha\beta\mu\nu} A^{b\alpha} A^{c\beta}\partial^\mu A^{a\nu} \quad (131)$$

$$-\frac{g^2}{2} f^{abc} f^{ade}\epsilon_{\alpha\beta\mu\nu} A^{b\mu} A^{c\nu} A^{d\alpha} A^{e\beta} \quad (132)$$

$$= -2\epsilon_{\alpha\beta\mu\nu}\partial^\mu A^{a\nu}\partial^\alpha A^{a\beta} + 2g f^{abc}\epsilon_{\alpha\beta\mu\nu} A^{b\alpha} A^{c\beta}\partial^\mu A^{a\nu} \quad (133)$$

$$= 2\partial^\mu\left[\frac{g}{3} f^{abc}\epsilon_{\alpha\beta\mu\nu} A^{b\alpha} A^{c\beta} A^{a\nu} - \epsilon_{\alpha\beta\mu\nu} A^{a\nu}\partial^\alpha A^{a\beta}\right] \quad (134)$$

Consider a term

$$\begin{aligned}
* &= f^{abc} f^{ade} \epsilon_{\alpha\beta\mu\nu} A^{b\mu} A^{c\nu} A^{d\alpha} A^{e\beta} \\
(a \leftrightarrow d) &\rightarrow -f^{ade} f^{dbc} \epsilon_{\alpha\beta\mu\nu} A^{b\mu} A^{c\nu} A^{a\alpha} A^{e\beta} \\
(a, \alpha \leftrightarrow b, \mu) &\rightarrow -f^{bde} f^{dac} \epsilon_{\mu\beta\alpha\nu} A^{a\alpha} A^{c\nu} A^{b\mu} A^{e\beta} \\
&= -f^{bde} f^{cad} \epsilon_{\alpha\beta\mu\nu} A^{a\alpha} A^{c\nu} A^{b\mu} A^{e\beta} \\
(c, \nu \leftrightarrow b, \mu) &\rightarrow -f^{cde} f^{bad} \epsilon_{\alpha\beta\nu\mu} A^{a\alpha} A^{b\mu} A^{c\nu} A^{e\beta} \\
&= -f^{cde} f^{abd} \epsilon_{\alpha\beta\mu\nu} A^{a\alpha} A^{b\mu} A^{c\nu} A^{e\beta}
\end{aligned}$$

Notice that

$$3* = (Red + Blue + Green)$$

On the other hand from Jacobi identity:

$$0 = f^{ade} f^{bcd} + f^{bde} f^{cad} + f^{cde} f^{abd}$$

hence $* = 0$ and the quartic term does not contribute to $\tilde{F}^{a}_{\alpha\beta} F^{a\alpha\beta}$.